

Five ways to look at a software portfolio

Growth–Share, Innovation Ambition, Ansoff, Wardley Mapping, Cost of Delay.
Practical decision frameworks for where the next unit of engineering capacity goes, and how to defend the answer.

5 models and the questions they answer

 Growth-Share

Who pays for whom?

A cash map. Which products fund the portfolio, and which are quietly consuming it.

 Innovation Ambition

How far from home are we betting?

The only allocation lens here. Splits the budget across core, adjacent and transformational work.

 Ansoff

Which direction does growth come from?

Four growth moves ranked by risk, so a diversification bet clears a higher bar than a pricing project.

 Wardley Mapping

Should this component exist at all?

Build, buy or rent, per component. The only lens that finds cost with no revenue line attached.

 Cost of Delay

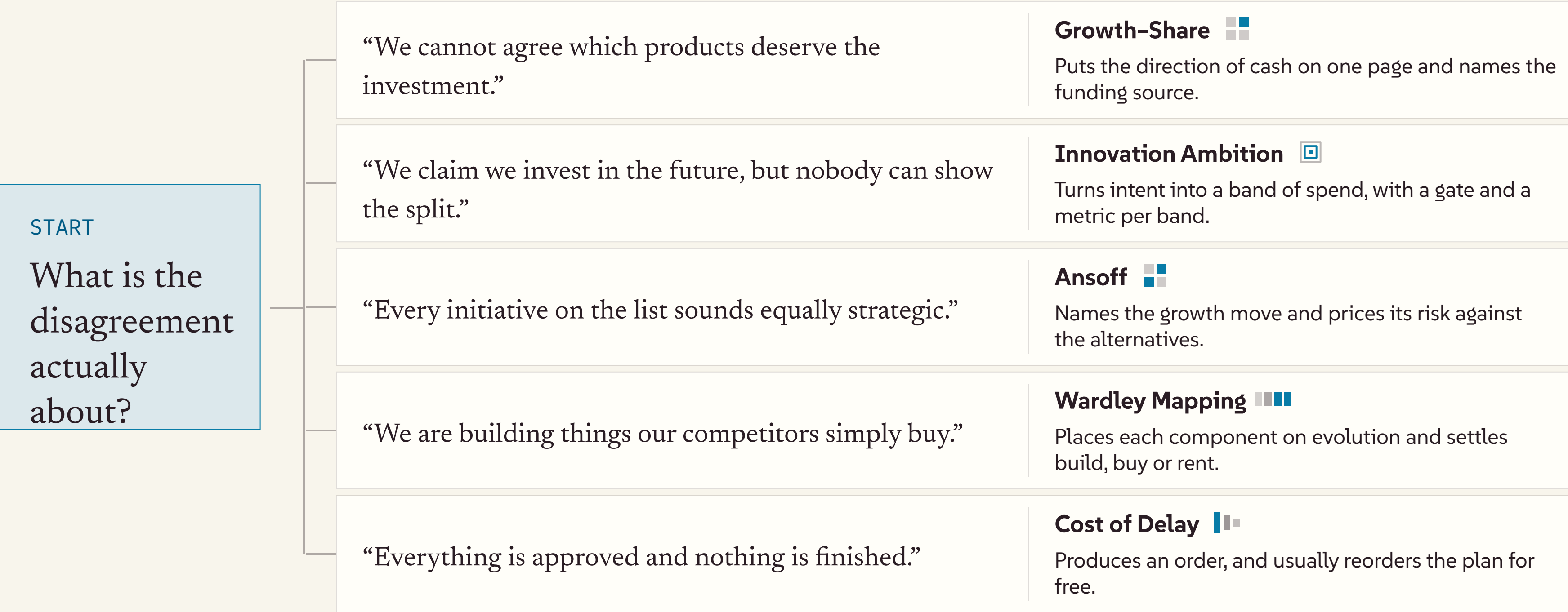
What order should we do it in?

Produces a ranked queue rather than a classification. Usually reorders the plan for free.

Cash reality constrains ambition, ambition picks the growth vector, the map decides what to build, and the queue decides what happens first.

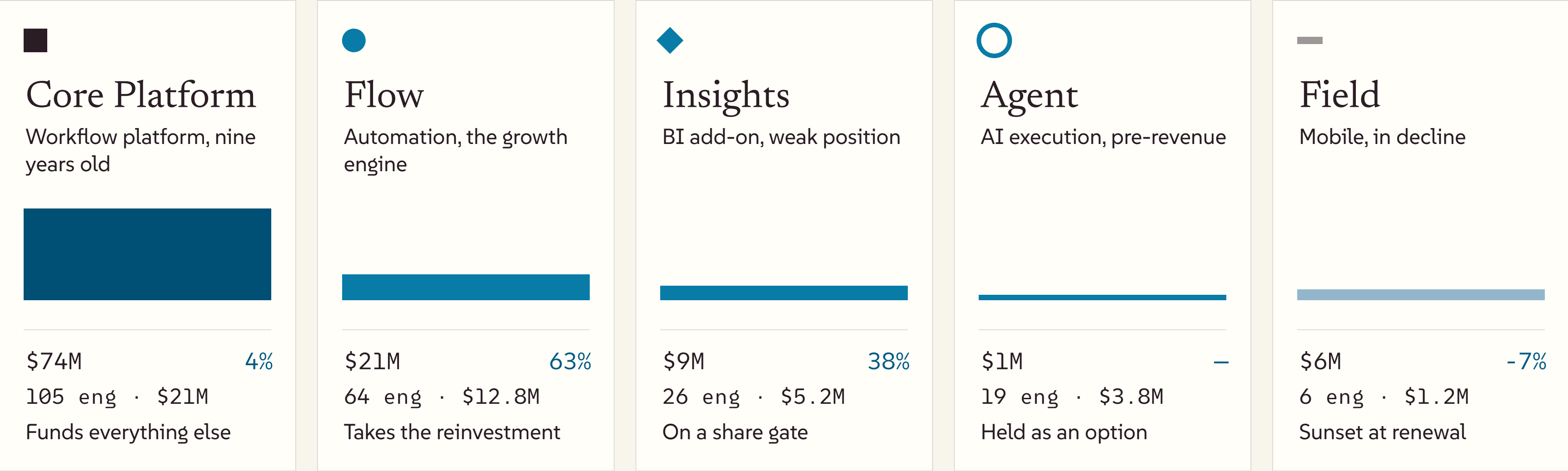
Start from the argument you are having

Each of the five settles a different dispute. Find the sentence you keep hearing and read across. If the argument is about one uncertain bet rather than the portfolio, none of them will settle it, so price it as an option instead.



Meet Arclight, the portfolio we will use throughout

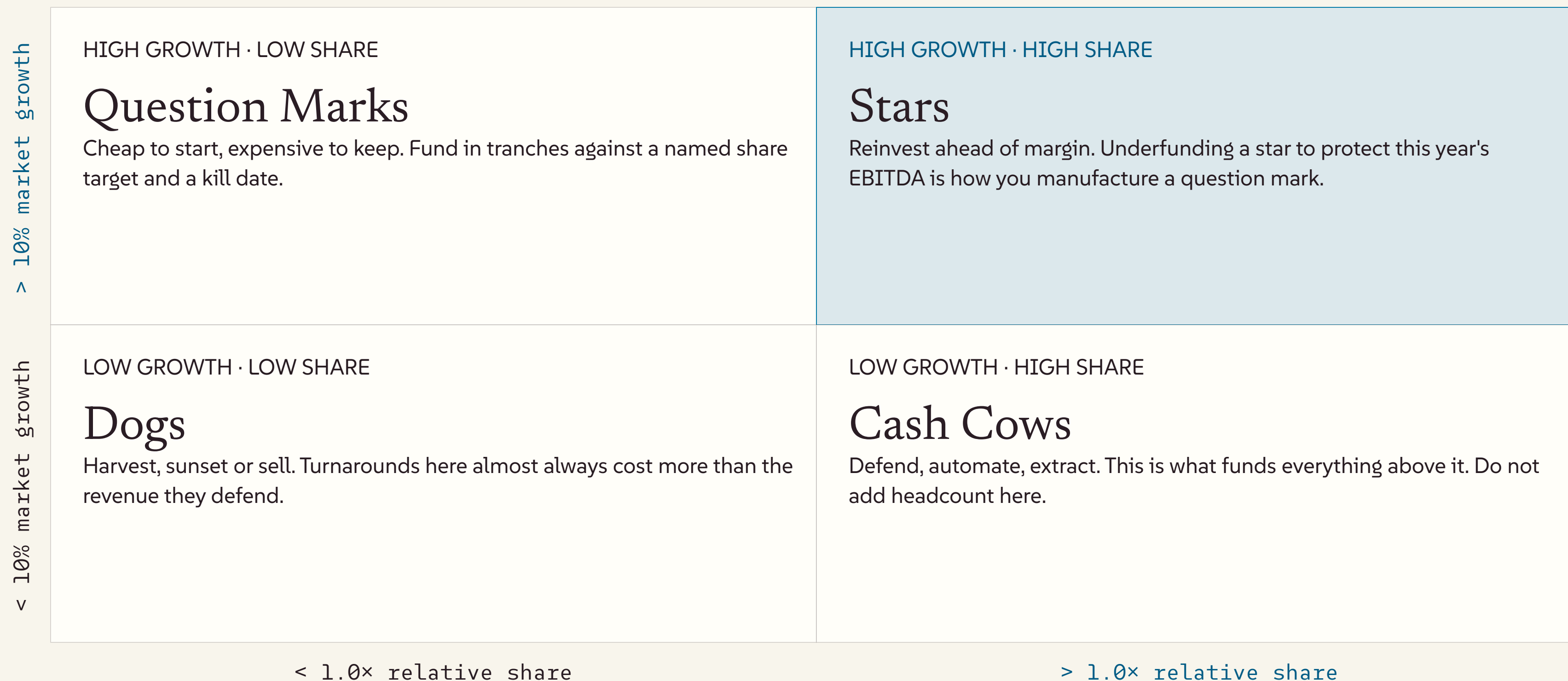
Arclight does not exist. It is an invented SaaS company, sized to be plausible. Its five products and their marks recur on every applied slide in this deck, so the five lenses can be compared on one portfolio rather than described in the abstract.



Bars show ARR to scale. Second line is headcount and annual engineering cost at \$200K fully loaded. Total \$111M ARR against \$44M of engineering, and Core Platform alone absorbs nearly half the team.

BCG Growth-Share Matrix

Two axes, and two thresholds that decide where each product lands: 10% market growth, 1.0× relative share. Everything else is judgment dressed as analysis.



Where Growth–Share came from, and how to build one

WHERE IT CAME FROM

Cash is the scarce resource

Henderson built the matrix on the experience curve: cumulative production volume lowers unit cost, so market share converts into margin. A portfolio should therefore be self-funding. The products with cost advantage in slow markets pay for the products fighting for position in fast ones.

The unit of analysis is a revenue-generating product in a defined segment, and the currency is cash rather than revenue or headcount.

HOW TO RUN IT

Five steps

- 01 Define the segment in writing, including who the largest competitor is.
- 02 $\text{Relative share} = \text{your revenue} \div \text{largest competitor's revenue in that segment}$.
- 03 $\text{Segment growth} = \text{YoY growth of the segment, not of your product}$.
- 04 Plot each product, bubble size = ARR, so scale is visible.
- 05 Apply the cutoffs (10% growth, 1.0× share) and read the quadrant.

HOW TO READ IT

Watch the direction of cash

The output is a direction of cash: cows fund stars and a small number of question marks; question marks are options with an expiry date; dogs release capacity rather than earn it.

The software caveat: share buys cost advantage through distribution, data and platform reuse rather than manufacturing volume. Net revenue retention and gross margin are the better local proxies for "cow", and a product with 0.3× share can still be structurally advantaged if it owns the workflow.

Arclight, a five-product SaaS portfolio

PRODUCT	ARR	GROWTH	SHARE	VERDICT
■ Core Platform	\$74M	4%	2.1×	Cash Cow
● Flow (automation)	\$21M	63%	1.4×	Star
◆ Insights (BI add-on)	\$9M	38%	0.3×	Question Mark
○ Agent (AI beta)	\$1M	—	0.1×	Question Mark
— Field (mobile)	\$6M	-7%	0.4×	Dog

Share here is relative to the largest competitor in each segment, rather than total market share.

THE REALLOCATION RULES

Send the cash upward

Core Platform's cash goes to Flow until Flow's growth drops below 25%.

Two question marks, one survives

Insights and Agent get 12 months and a 0.8× share gate. Whichever misses it is folded into Flow.

Dogs get a date

Field enters maintenance now, sunset announced at renewal. Its six engineers move to Flow.

Three ways this matrix lies to you

01

Market share is a guess

In software, segment boundaries are drawn by whoever wants the budget. Define the segment before you measure share, and write the definition down.

02

It ignores what products share

A dog that carries your identity layer, your data model or your enterprise contracts is not a dog. Score dependencies separately.

03

Blind to option value

The matrix has no opinion on option value. Early bets that could reset the category always look like question marks worth killing.

USE IT WHEN

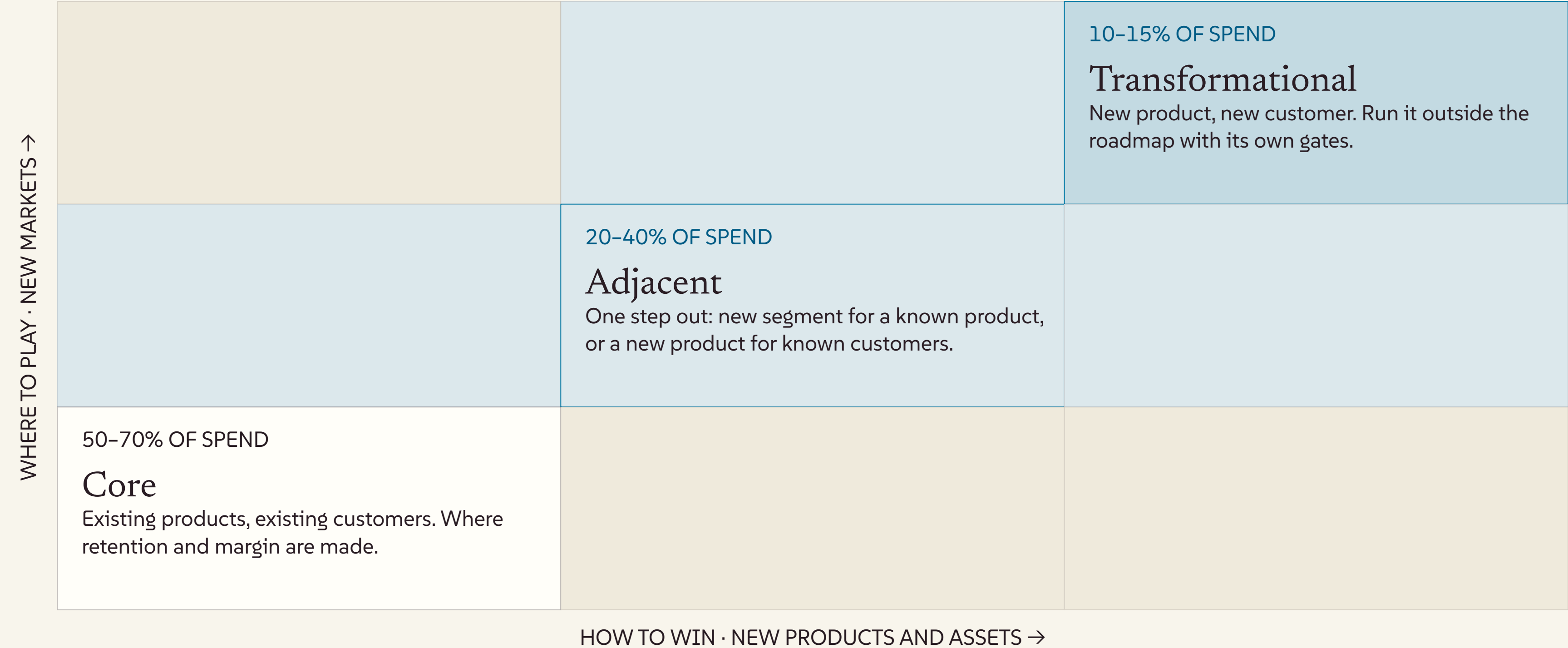
You have four or more revenue-generating products and a capacity fight to settle.

DON'T

Single-product companies, pre-revenue bets, or anything you'd defend on strategic-option grounds.

Distance from today's products and today's customers. Nagji and Tuff's 2012 average was 70/20/10; their own sector split put technology firms nearer 45/40/15. For software the defensible band is wider.

Innovation Ambition Matrix



What the ambition split is actually claiming

WHERE IT CAME FROM

Ambition levels need different rules

Innovation fails as a budget line because core work and distant bets compete for the same money under the same metrics, and the core always wins, because its returns are nearer and better evidenced.

Nagji and Tuff's fix is to classify every investment by its distance from today's products and today's customers, then govern each band differently: different funding gates, different metrics, different tolerance for failure.

HOW TO RUN IT

Five steps

- 01 List every funded investment, including the ones nobody calls innovation.
- 02 Classify each on both axes: how new is the customer, how new is the product or asset.
- 03 Sum actual spend per band. Engineers, not intentions.
- 04 Compare the real split against your stated ratio.
- 05 Set metrics per band and a gate per band, then ring-fence the teams.

HOW TO READ IT

The gap is the finding

The output is an allocation and, more usefully, the distance between the allocation you announced and the one your timesheets describe. Most portfolios discover they are running 90/10/0.

The 2012 study found 70/20/10 among outperformers, with a roughly inverted return profile. Treat it as a benchmark. Technology firms in that same study sat nearer 45/40/15. Mature category under attack: push transformational up. Fast-growing core with real retention: 80/15/5 is defensible.

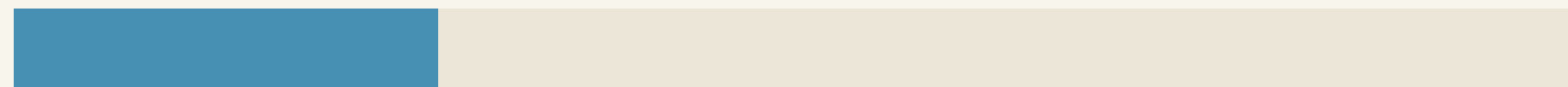
Arclight's engineering budget, split three ways

■● Core · Platform + Flow reliability 64%



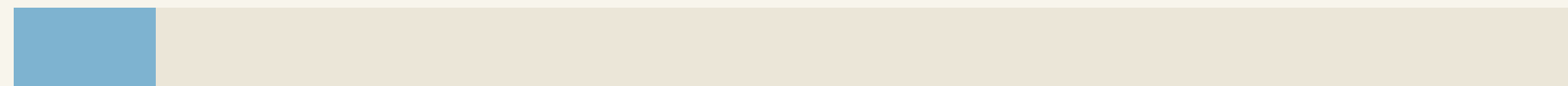
142 of 220 engineers. Retention work, migration debt, enterprise controls.

●◆ Adjacent · Flow for mid-market, Insights 27%



59 engineers. Same product, new segment; new product, same buyer.

○ Transformational · Agent 9%



19 engineers, separate P&L, milestone funding, no roadmap obligations. Deliberately below the band, and the reason is on the record.

The drift check: at the last review, 18 of the 19 Agent engineers were still on Agent. That number is the only real evidence the split exists.

RULES THAT MAKE THE SPLIT REAL

Separate the funding gates

Core is annual and headcount-based. Transformational is milestone-based and can be stopped mid-quarter.

Different metrics per band

Core is judged on NRR and margin. Transformational is judged on learning velocity and evidence, not revenue.

Shift the ratio deliberately

Mature category under attack: push to 50/30/20. Fast-growing core: 80/15/5 is defensible. Name the number and the reason.

Governance kills the transformational band

01

The split becomes a slogan

Announced in January, quietly 95/5/0 by March as core incidents pull people back. Ring-fence the teams or drop the claim.

02

Relabelling instead of reallocating

Core roadmap items get renamed "transformational" to look ambitious. Test it: would this ship to a customer you have today?

03

Core work looks unglamorous

The core band is where the money is. Staff it with your strongest people, or the funding for everything else disappears.

USE IT WHEN

Setting an annual budget, or explaining why a bet gets funded on different terms.

DON'T

Use it to rank individual features. It works on portfolios of investment, not backlogs.

Ansoff Growth Matrix

Four growth moves. The risk multipliers are the useful part.

MARKETS · EXISTING → NEW

NEW MARKET · EXISTING PRODUCT

4× risk

Market development

New geography, new segment, new channel. Cheapest real growth if the product genuinely transfers.

NEW MARKET · NEW PRODUCT

8× risk

Diversification

Two unknowns at once. Justified only by a real asset advantage, or when the core is structurally dying.

EXISTING MARKET · EXISTING PRODUCT

1× risk

Market penetration

Pricing, packaging, activation, win rate. Exhaust this before anything else on the board.

EXISTING MARKET · NEW PRODUCT

2× risk

Product development

Sell more to buyers you already have. Fastest compounding move in most SaaS portfolios.

PRODUCTS · EXISTING → NEW

The oldest of the five, and the simplest to abuse

WHERE IT CAME FROM

Growth has only four directions

Ansoff's argument was that any growth initiative is some combination of product novelty and market novelty, and that each kind of novelty introduces its own class of unknown: a technical one, a demand one, or both at once.

The value is in naming which unknowns you have taken on, so two initiatives asking for the same money can be compared on more than their forecast.

HOW TO RUN IT

Five steps

- 01 State the initiative as one sentence: sell X to Y.
- 02 Classify X and Y as existing or new using the buyer test. A different buyer or a different budget line means new.
- 03 Assign the risk multiplier: penetration 1×, product development 2×, market development 4×, diversification 8×.
- 04 Require the expected return to clear that multiplier, not a flat hurdle rate.
- 05 Sequence the cheapest unknowns first, and hold the 8× move as an option.

HOW TO READ IT

Use it to set the bar

Used well, Ansoff is a sequencing tool: it forces penetration work to be exhausted before new products are funded, and it prices diversification honestly against the alternatives.

The multipliers are heuristics, widely quoted and not empirically derived. Their job is discipline. A diversification bet has to clear a visibly higher bar than a pricing project of the same size. Treat them as a ratio between options, never as an expected loss.

Sequencing Arclight's next \$40M

● Q1-Q2	Penetration	Re-price Flow by usage tier, fix trial-to-paid activation, close the win-rate gap on two named competitors.	\$6M · 1× risk
◆ Q2-Q4	Product development	Ship Insights as a paid module to the installed base. Known buyer, known problem, new SKU.	\$14M · 2× risk
● Q3-Q4+	Market development	Take Flow into mid-market through partners. Same product, new buying process, new pricing floor.	\$15M · 4× risk
○ Optioned	Diversification	Agent for a market Arclight does not serve. Held at \$5M and milestone-gated, not scaled.	\$5M · 8× risk

The multipliers are not precise. They exist so a \$5M diversification bet has to clear a far higher bar than a \$5M pricing project.

It names the move. It will not make the call

01

"New" is doing a lot of work

A new tier for existing buyers is not a new product. Define new as a different buyer or a different budget line.

02

Nothing about capacity

The matrix will happily suggest four moves you cannot staff. Pair every quadrant with a named team, or it is a wish list.

03

Penetration gets skipped

It is the least exciting box and usually the highest return. Most portfolios have a pricing or activation project worth more than their next new product.

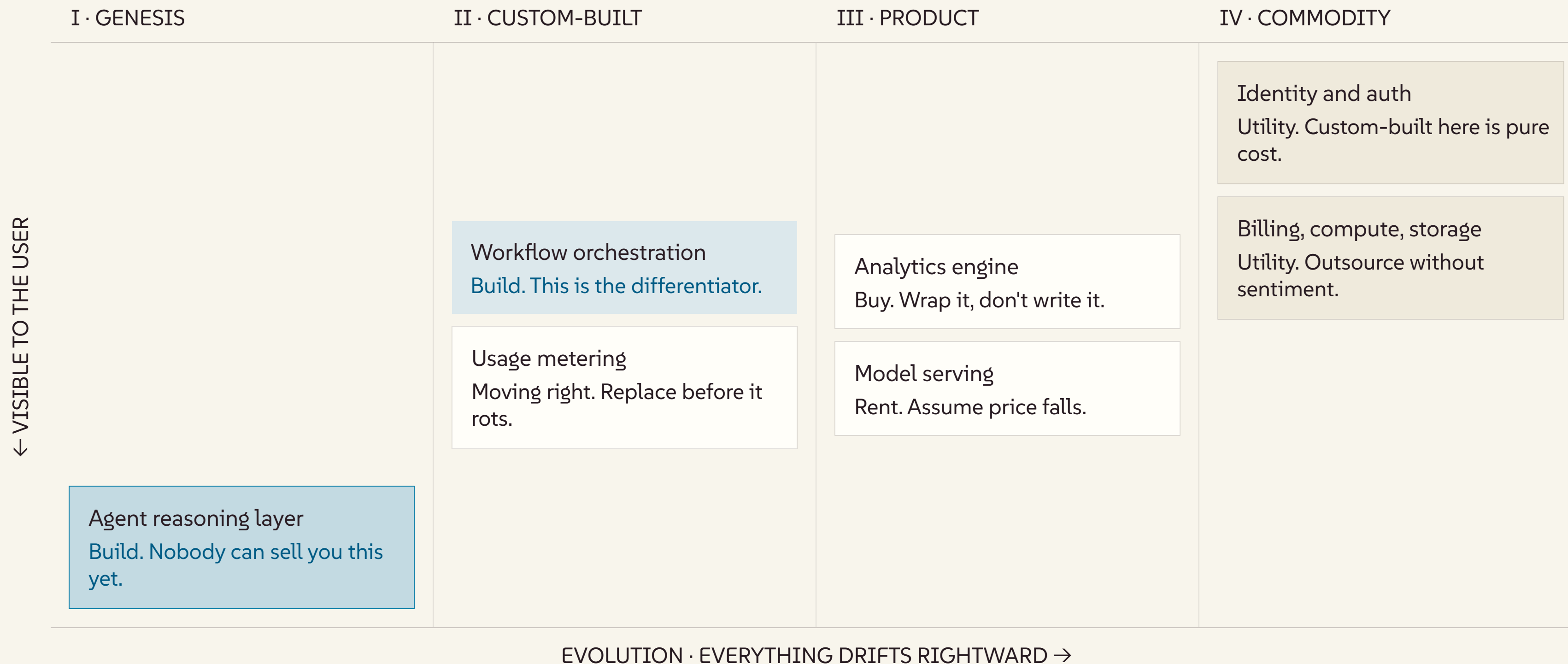
USE IT WHEN

Sequencing growth initiatives and pricing their risk against each other.

DON'T

Treat it as a portfolio review. It says nothing about what to stop doing.

Wardley Mapping



Position, movement, and what to stop building

WHERE IT CAME FROM

Everything drifts toward commodity

A map starts from a user need and chains downward through every component that need depends on. Each component is then placed on an evolution axis by its characteristics: how uncertain, how differentiating, how widely available.

The claim is that supply and demand competition pushes every component rightward over time, and that the correct practice for a component depends entirely on where it currently sits, rather than on what it was when you first built it.

HOW TO RUN IT

Five steps

- 01 Write the user need at the top. Not a feature, a need.
- 02 Chain the components beneath it until you reach things you rent.
- 03 Place each component on the evolution axis: genesis, custom-built, product, or commodity.
- 04 Mark the direction of travel, and mark where you have inertia: sunk cost, a team, a contract.
- 05 Apply doctrine per stage: build genesis, build custom where you differentiate, buy product, rent commodity.

HOW TO READ IT

The uncomfortable column

Two errors show up on almost every map. Custom-building something that is already a commodity, which is invisible cost. And buying something in genesis that is your actual differentiator, which is outsourced strategy.

The output is shared situational awareness rather than a priority list, and the argument about placement is where most of the value sits. Redraw it when the market moves, not on a schedule.

What the map told Arclight to stop doing

■ DECISION 01

Kill the in-house identity service

Nine years old, four engineers, zero differentiation. It sits in the commodity column and is bought by everyone else in the category. Migrate to a provider and redeploy the team to Flow.

Inertia: it was someone's promotion.

● DECISION 02

Stop shopping for a workflow engine

Orchestration is the one custom-built component that customers choose Arclight for. Buying it would hand the differentiator to a vendor who also sells to competitors.

This reverses a procurement decision already in flight.

○ DECISION 03

Rent model serving, own the reasoning layer

Inference is commoditising and its price is falling; the domain reasoning on top of it is genesis and nobody sells it. Build one, rent the other, and expect to renegotiate annually.

Revisit when serving prices halve again.

None of the other frameworks in this deck would have surfaced the identity service. It has no revenue line, so it never appears as a product.

A map nobody redraws is a poster

01

Placement becomes the meeting

Teams spend two hours arguing whether something is product or commodity. Timebox it: the column matters, the exact position within it does not.

02

It needs a facilitator

Unlike a 2×2, a map is hard to produce well on a first attempt. Expect the first two to be wrong and treat them as practice.

03

It will not prioritise for you

A map tells you what to build, buy or rent. It will not rank your initiatives or size your bets, so pair it with something that does.

USE IT WHEN

You are deciding build versus buy, or you suspect you are custom-building commodities.

DON'T

You need a funding split or a ranked backlog. The map has no opinion on either.

Cost of Delay and WSJF

THE RATIO

Weighted Shortest Job First = Cost of Delay ÷ Job Duration

Cost of Delay is what one week of waiting costs you. Duration is how long the work takes. Rank by the ratio, take from the top, re-rank next cycle.

COMPONENT 01

User and business value

Revenue protected or gained per week. The only component you can sometimes state in currency.

COMPONENT 02

Time criticality

Does the value decay? A renewal deadline, a competitor launch or a regulatory date all make waiting expensive.

COMPONENT 03

Risk and opportunity enablement

What this unblocks or de-risks. Where platform work earns its ranking instead of pleading for it.

Queues are the cost nobody budgets

WHERE IT CAME FROM

Time is the constraint

Without a cost of delay, an organisation optimises for what it can see: utilisation, headcount efficiency, cost per feature. Queues are invisible on every dashboard, so the economic cost of work sitting still never enters a decision.

Putting a number on a week of delay changes the argument from which initiative is most important to which order costs us least. That reordering is usually free.

HOW TO RUN IT

Five steps

- 01 Estimate cost of delay per week for each initiative. In currency if your economics allow, otherwise on a relative 1-2-3-5-8-13 scale.
- 02 Estimate duration, not effort. A six-week job with two people beats a two-week job that waits a quarter for a dependency.
- 03 Divide to get the ratio, and sort.
- 04 Take work from the top only, and resist splitting capacity across the whole list.
- 05 Re-rank every cycle. The ranking is a snapshot of economics, and economics move.

HOW TO READ IT

Short and valuable first

The counter-intuitive result is that a small, cheap, moderately valuable job frequently outranks a large, exciting, highly valuable one. That is the point of it.

Absolute scores mean nothing. Only the ordering does. And the ranking assumes you can actually start the top item. It says nothing about whether the right team is free, which is why it pairs with a capacity view rather than replacing one.

The pricing project outranks the new module

INITIATIVE	COD / WEEK	DURATION	WSJF	RANK
● Flow usage-tier re-pricing	\$180K	8 weeks	22.5	1
● Trial-to-paid activation fixes	\$95K	6 weeks	15.8	2
■ SOC 2 controls for enterprise deals	\$210K	16 weeks	13.1	3
◆ Insights paid module	\$240K	34 weeks	7.1	4
— Field mobile refresh	\$20K	20 weeks	1.0	5

Insights has the highest cost of delay on the list and still ranks fourth. Duration is doing the work, and it is the number most worth attacking.

Field ranks last on economics and last on the growth-share read. Two independent frameworks agreeing is the strongest signal in the deck.

Numbers that only look economic

01

Relative scales get gamed

Once teams learn that 13s win, everything becomes a 13. Calibrate against two or three reference initiatives everyone already agrees on.

02

Duration is the weak input

People estimate effort and call it duration, ignoring the queue time that dominates. Measure real cycle time from finished work and use that instead.

03

A ranking is not a plan

WSJF assumes the top item can start now. Without a capacity view it produces an order nobody can execute, and the list quietly gets ignored.

USE IT WHEN

You have more initiatives than capacity and the argument has stalled on importance.

DON'T

Use it to compare a discovery bet against a delivery project. Options and deliverables do not share a denominator.

Timothy Luehrman, HBR 1998, applied to product bets. For a high-variance idea, commit a small amount and decide later.

Real options and staged gates

TRANCHE 0 Discovery

Is the problem real, and does anyone pay to solve it today?

Kill if fewer than five of twenty target accounts describe the problem unprompted.

\$0.4M · 8 weeks

TRANCHE 1 Prototype

Can we build it well enough to be useful with the assets we already have?

Kill if accuracy stays below the threshold a human reviewer would accept.

\$1.5M · 2
quarters

TRANCHE 2 Design partners

Will someone pay, and does the value survive contact with their data?

Kill unless two partners convert to paid at target price.

\$5M · 2
quarters

TRANCHE 3 Exercise

Scale funding, a real P&L, and a place in the portfolio review.

This is the only tranche that buys revenue. The first three buy information.

Full funding

Agent, priced as an option

WHAT THE OTHER FRAMEWORKS SAID

Growth–Share: kill it

\$1M ARR, 0.1× relative share, no measurable segment growth. A question mark that fails its own gate.

Ansoff: 8× risk

New product, new market. The highest bar in the matrix, and no forecast clears it honestly.

As an option: \$1.9M to find out

Tranches 0 and 1 cost less than three months of Core Platform maintenance, and resolve the two questions the forecast was guessing at.

THE TERMS, WRITTEN DOWN

UNCERTAINTY

Will operations teams delegate multi-step work to software they cannot fully inspect?

CAP

\$5M total across three tranches, 19 engineers, no claim on the Flow roadmap and no shared on-call.

KILL CRITERIA

Signed by CTO and CFO before tranche 0 opens. A missed criterion stops the work, without a review meeting.

EXPECTED OUTCOME

Most likely this dies at tranche 2 having cost \$1.9M and answered the question. That counts as the option working.

Five lenses, five different jobs

FRAMEWORK	AXES OR INPUTS	UNIT OF ANALYSIS	OUTPUT	BLIND SPOT
 Growth-Share	Market growth × relative share	Revenue-generating product	Cash flow direction	Shared assets, option value
 Innovation Ambition	Where to play × how to win	Investment programme	Budget split across three bands	Silent on which market
 Ansoff	Products × markets	Growth initiative	Move ranked by risk	Capacity, and what to stop
 Wardley Mapping	Value chain × evolution	Component or capability	Build, buy or rent	No ranking or sizing
 Cost of Delay	Delay cost ÷ duration	Initiative in a queue	A ranked order	Ignores capacity; duration is guessed

None of the five has an opinion on option value. Slides 25 and 26 close that gap.

Run it on a cadence, or don't run it

ANNUAL

Set the ambition split

Decide the core / adjacent / transformational ratio and ring-fence the teams.

SEMI-ANNUAL

Redraw the map

Wardley, from the current user need. Check what has drifted right since last time.

QUARTERLY

Check the cash map

Growth and share by product. Confirm the cows still fund the stars.

PER INITIATIVE

Name the growth move

Every funded initiative declares its Ansoff quadrant and clears that bar.

None of these five tell you what to do. They make disagreement specific, which is most of the work.